

Marketing 101 for Engineers

Aditi Agarwal

Introductions First

2012



Aditi Agarwal
Final Year B.Tech
Dyes

14 Years

2026



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Indegene

How our day together will look like

1. Marketing 101

2. Brand

3. Sales & Marketing

4. Value to us Engineers

What's Marketing



What do you think is the cost of this cup of coffee



Does the price change now ?

The coffee doesn't really change, the **experience** changed



1500 INR



8000 INR

Both protect your feet , but **identity** changed



20 INR



120 INR

Its all H2O, **perception** changes



Coke has >50% MS with \$40-45 B Revenue
Pepsi has 20% with \$20-25B

Derive a premium price, loyalty, customer base and business continuity

Marketing is the process of creating, communicating,
and capturing value for a product/people

“Good marketing increases the value customers perceive — and that allows companies to charge more.”

Lets break down the process of marketing with a story



Martin Eberhard
Mark Tarpenning

Tesla Roadster (2008)



Tesla Model S (2012)



Tesla Model X (2015)



Tesla Model 3 (2017)



Tesla Cybertruck (2023)



Key innovations

- First highway-capable electric sports car
- ~393 km range (unheard of for Evs then)
- Lithium-ion battery pack using thousands of laptop cells
- 0–60 mph in ~4 seconds

Purpose

- Prove EVs can be fast, desirable & viable

Limitations

- Extremely expensive (~\$100k+)
- Small production volume (~2,450 units)

Strategic learning

- Battery management
- Electric drivetrain engineering
- Early Tesla software systems
- Eco Crash

Major improvements

- Range increased to 426+ km
- Large 17-inch touchscreen interface
- OTA (over-the-air) software updates
- Much better battery architecture

Breakthrough features

- Autopilot hardware (2014)
- Software-defined car
- Supercharger network launched alongside

Impact

- Won Motor Trend Car of the Year (2013)
- First EV to compete directly with luxury sedans

Strategic shift

Tesla moved from experiment → real car company
Wall street ,

New improvements

- Falcon wing doors
- Massive panoramic windshield
- 7-seat configuration
- Bioweapon defence HEPA filter
- Improved autopilot hardware

Performance

- One of the fastest SUVs in the world
- 0–60 mph ~3.2 sec (Performance version)

Challenges

- Extremely complex engineering
- Production delays (“production hell” early signs)

Strategic role

Expand EVs into the SUV segment.

Major changes

- Price target: ~\$35,000
- Extreme minimalist interior
- Single center screen (no instrument cluster)
- Huge manufacturing scale
- New battery cost structure
- Technology improvements
- More efficient powertrain
- Simplified design
- Advanced Autopilot hardware

Impact

- One of the best-selling EVs
- Production scaled to hundreds of thousands

Strategic importance

This was Tesla’s iPhone moment.

Major innovations

- Stainless steel exoskeleton
- 48-volt electrical architecture
- Steer-by-wire
- Gigacasting
- Structural battery pack

Performance

- 0–60 mph ~2.6 sec (Cyberbeast)
- ~500+ mile range target variants

Strategic goal

Enter the U.S. pickup market, the most profitable vehicle segment.

Today, EV is almost becoming mainstream but Tesla Remains ahead



Lets break down the process of marketing with a story

Tesla didn't randomly build an electric car and hope people would buy it — it studied the entire system first.

Who are you?

Its about how you see yourself as a company. Tesla said its

technology and innovation company that builds cars.

Key values communicated:

- innovation
- sustainability
- futuristic technology
- software-driven mobility

Who will buy?

Tesla initially targeted early adopters and premium buyers, not the mass market.

Key customer segments:

- technology enthusiasts
- environmentally conscious consumers

- luxury car buyers
- performance seekers

Later expansion:

- mainstream middle-class buyers (with Model 3 and Model Y)

Who were they pitched against?

Tesla positioned itself against luxury and performance brands, not economy cars.

Key competitors initially:

- BMW
- Mercedes-Benz
- Audi

Later competitors:

- traditional EV manufacturers
- BYD
- Rivian
- Lucid

Strategic positioning:

Tesla said:

“We are not another car company — we are the future of mobility.”

Who Helped?

Tesla's success also depended on important collaborators .

Key collaborators:

- battery suppliers (Panasonic, CATL)
- government EV subsidies and incentives
- charging infrastructure partners
- software and semiconductor suppliers

Tesla also built its own ecosystem:

- Supercharger network
- Gigafactories

What Worked ?

Tesla watched the market shift or context of market

Climate and sustainability concerns

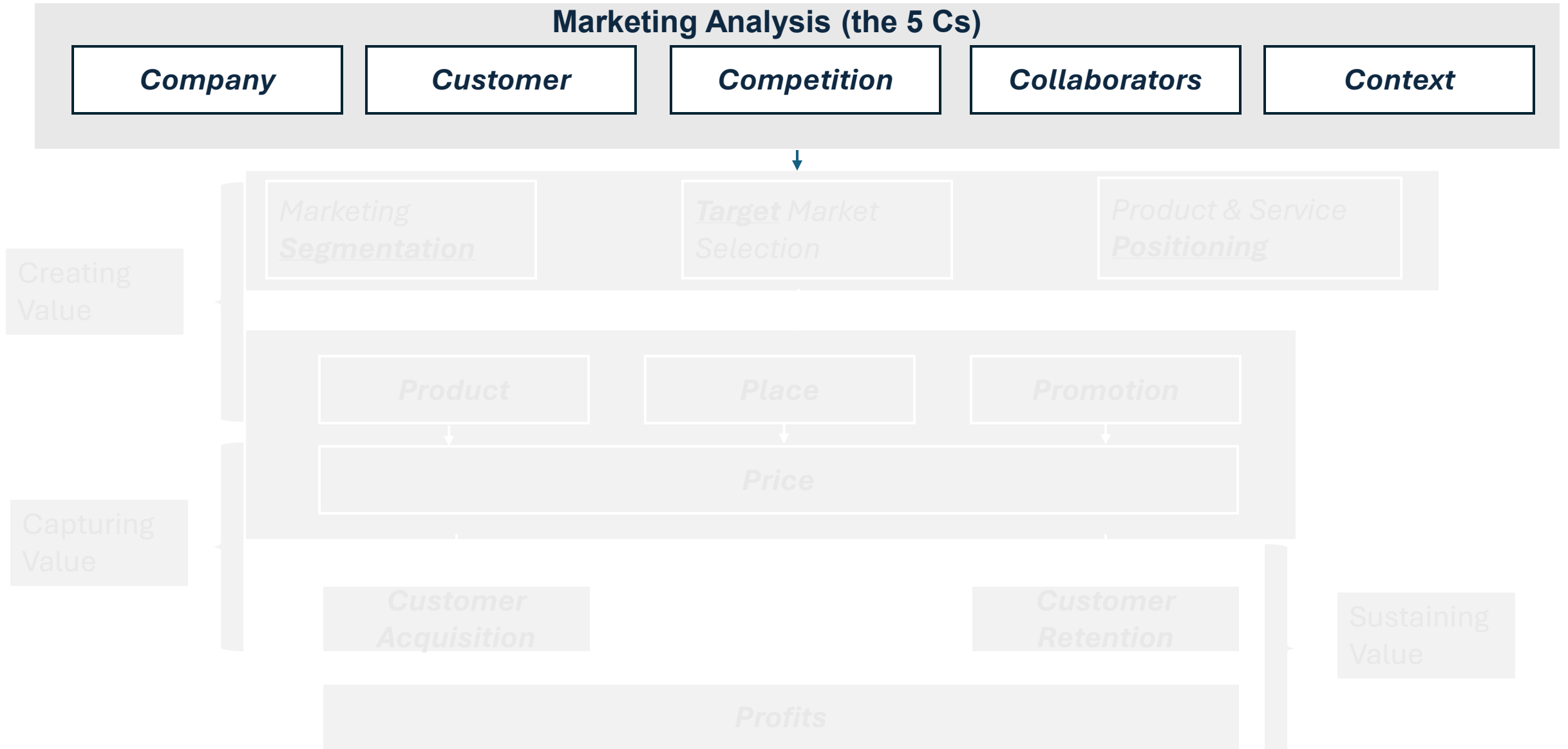
- Rising awareness of climate change
- Global pressure to reduce carbon emissions
- Consumers becoming more environmentally conscious

Government policies

- EV subsidies and tax incentives
- Stricter emission regulations for car manufacturers
- Long-term plans to phase out gasoline vehicles in many countries

These 5 C – Company, Customer, Competition, Collaborators and Context

Summarizing Marketing 101



My own Marketing to land my first job

C	My Situation (2012)
Company/ Me	Chemical Technologist. Sound Organic Chemistry knowledge. Specialty Dyes and Pigments. Strong analytical thinking. Curiosity about business. Good at talking & communication
Customers	Hiring managers looking for engineers who can understand chemistry and sell
Competitors	Peers. Lateral Hiring Community. B School Grads. Other GTE
Collaborators	Professors, Mentors, Recruiters, Alumni networks
Context	Globalization and increasing demand for technically sound salespeople who can interact with international clients and be 2 in a box solution . World was recovering from 2008 crisis

“The 5Cs help a company understand the market environment.

But understanding the market is not enough.

The next challenge is deciding **which customers to focus on and how to position the product for them.**”

Starting to Action

“Does everyone in the world want the same type of car?”

Demography

- economy buyers
- mid-range buyers
- luxury buyers

Lifestyle

- environmentally conscious consumers
- technology enthusiasts
- performance lovers
- Status Lovers

Age

- young professionals
- families with children
- older buyers

Geography

- Urban
- Rural

*Tesla didn't try to sell electric cars to everyone at the beginning
It first sold them to the people most likely to believe in the future*

- luxury buyers- high income
- Tech Focused
- Sustainable and innovation
- Early Adopters
- Urban

How should this target list perceive Tesla

- slow
- unattractive
- low performance
- boring
- limited range

Technology Leadership

Tesla cars were marketed as:

- software-driven vehicles
- constantly improving through updates
- technologically superior

Tesla emphasized acceleration and speed.

Where Tesla then stood

Brand	Positioning
BMW	Driving performance
Mercedes	Luxury and comfort
Toyota	Reliability
Tesla	Future technology and sustainability

Before Tesla:

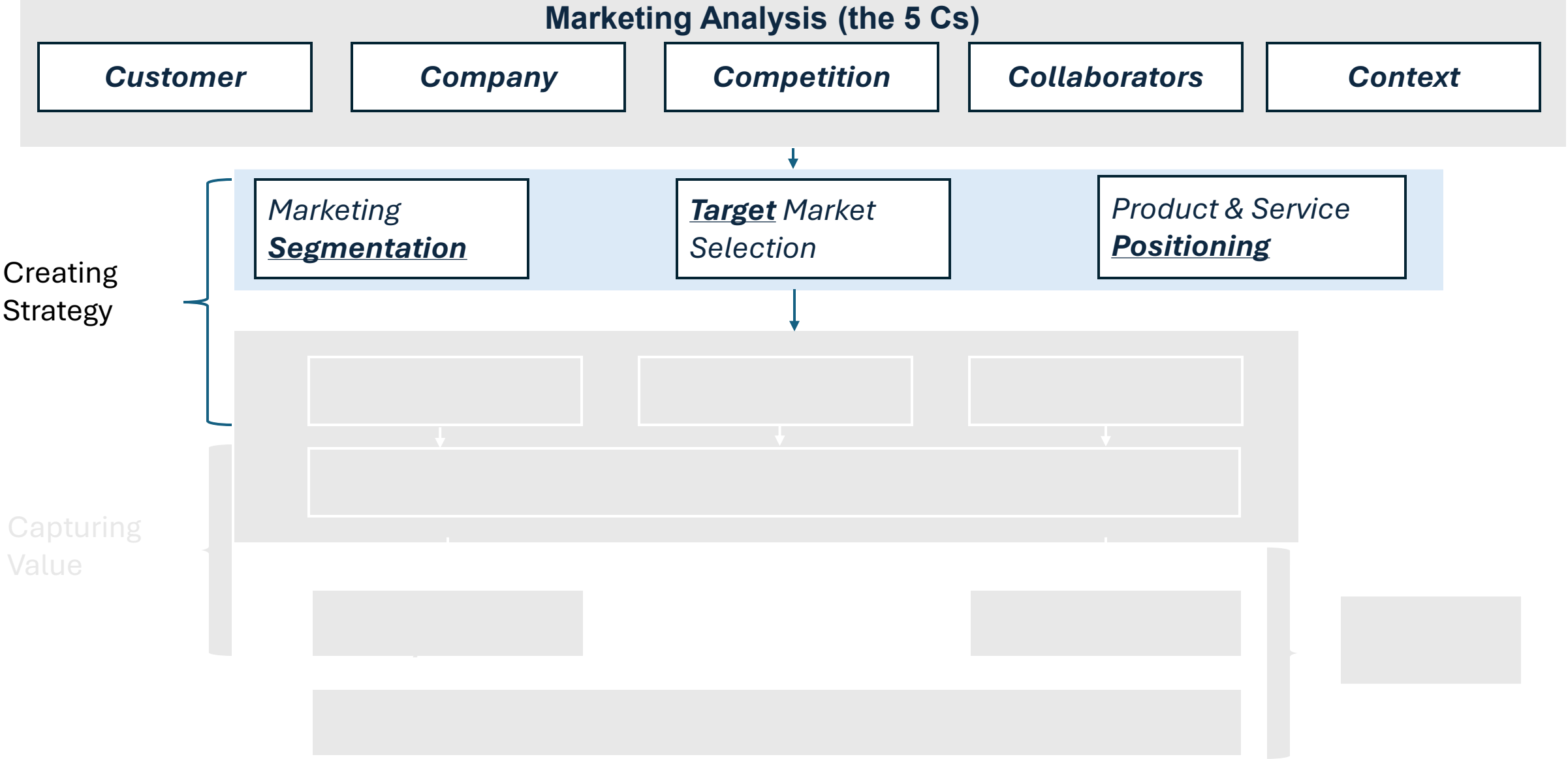
EV = compromise.

After Tesla:

EV = innovation and performance.

*“Tesla didn’t just build an electric car.
It positioned electric cars as **the future of transportation.**”*

Summarizing Marketing 101



Lets take few more examples for positioning

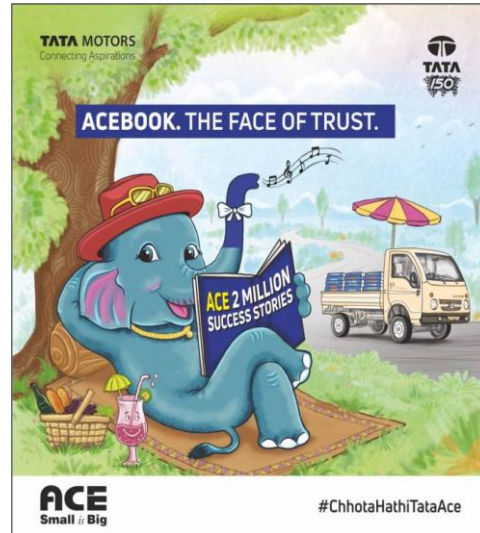
Attribute Positioning



Castrol = **strong engine protection**

Dettol didn't just sell antiseptic liquid.
It sold **certainly against invisible threats**

Benefit Positioning



This vehicle helps **people earn a livelihood**



Maruti = **great mileage and low running cost**

Application Positioning



Customers associate Mahindra Jeeps with **durability and off-road capability.**

Emotional Positioning



Asian Paints positions itself as **a brand that helps create beautiful homes and memories**

No more tears
from 'soap in the eyes'



Targets the mothers

Positioning Statement

For

(target audience)

Our Product/Brand is

(competitive frame/ category)

That provides

(single most important benefit)

because

(reason to believe)

For

Consumers who look at 4 wheeler as a means
of enhancing life

(target audience)

Our Product/Brand is

Tata Ace is a minitruck

(competitive frame/ category)

That provides

Makes you feel good about life and
yourself

(single most important benefit)

because

because we help you make more
money with every trip along with making you
comfortable while driving

(reason to believe)



Dove Brand Positioning

For Independent and Confident women

Dove is a moisturizing soap

that has 1/4th moisturizing cream

which leaves skin softer and younger looking

For

companies selling complex chemicals and products to global customers
(target audience)

Our Product/Brand is (I am a)

a young , curious & enthusiastic chemical technologist who who has the ability of clear communication
(competitive frame / category)

That provides

the ability for the company to understand and cater their customers in a techno commercial manner with one single employee instead of two
(single most important benefit)

because

of strong chemical training combined with curiosity about customer problems coupled with sound business understanding
(reason to believe)

My own Marketing to land my first job

Stage	Key Question	My Approach
Segmentation	What career paths exist for me?	1. Technical Sales 2. Technical Services 3. Field Sales 4. Consulting
Targeting	Which segment should I focus on?	1. Technical Sales 2. Technical Services
Positioning	How should employers perceive me?	An engineer who can bridge technology and customers , translating unmet customer need into business value

One you have targeted and positioned...

You have to communicate the positioning to the audience you have targeted ?

How do you buy ?????
Lets try buying something from our favourite “Apni Dukan”

Product

- electric cars
- software-driven vehicles
- autopilot
- futuristic design

Price

- premium pricing initially
- signals advanced technology

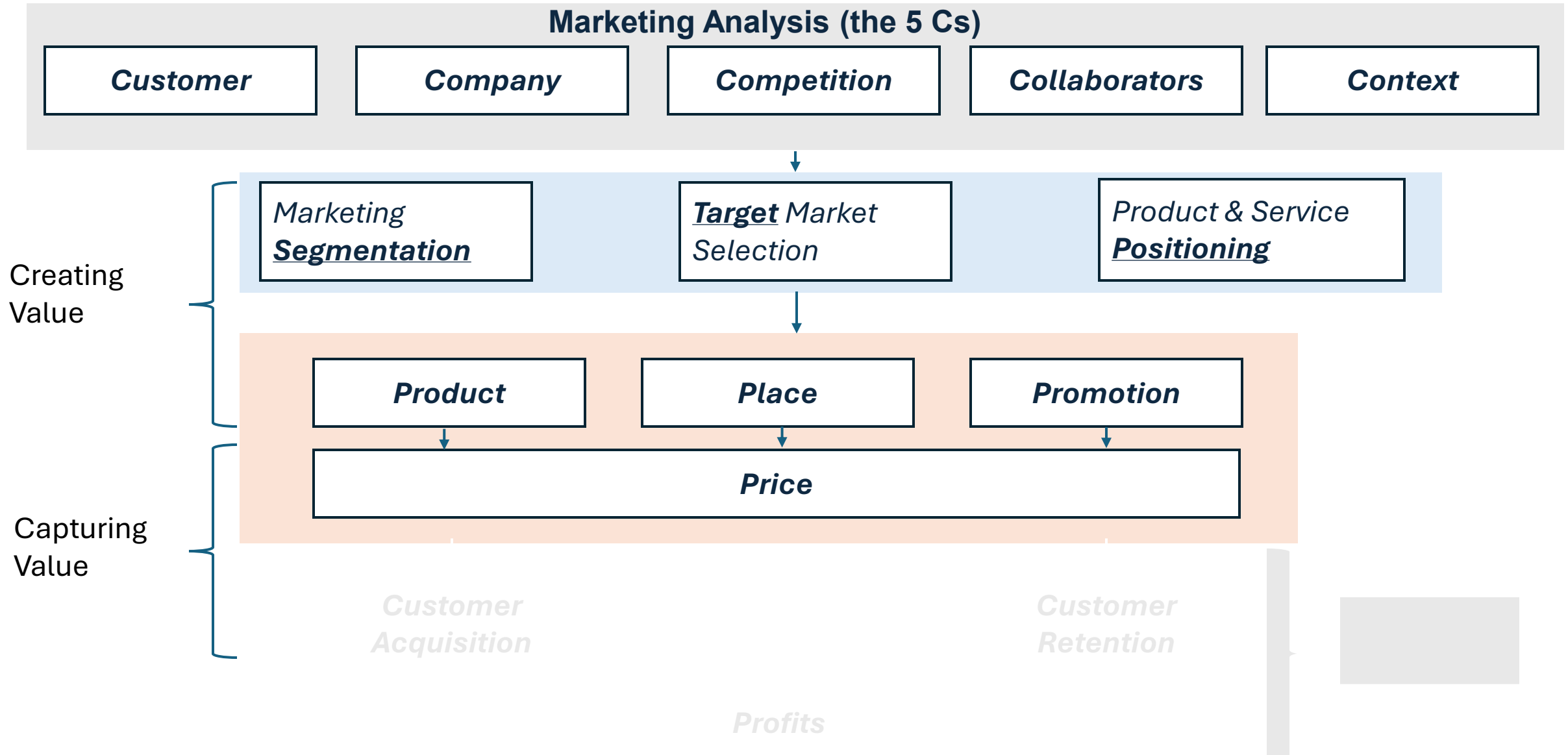
Place

- direct online sales
- Tesla experience centers

Promotion

- product launches
- Elon Musk visibility
- minimal traditional advertising

Summarizing Marketing 101

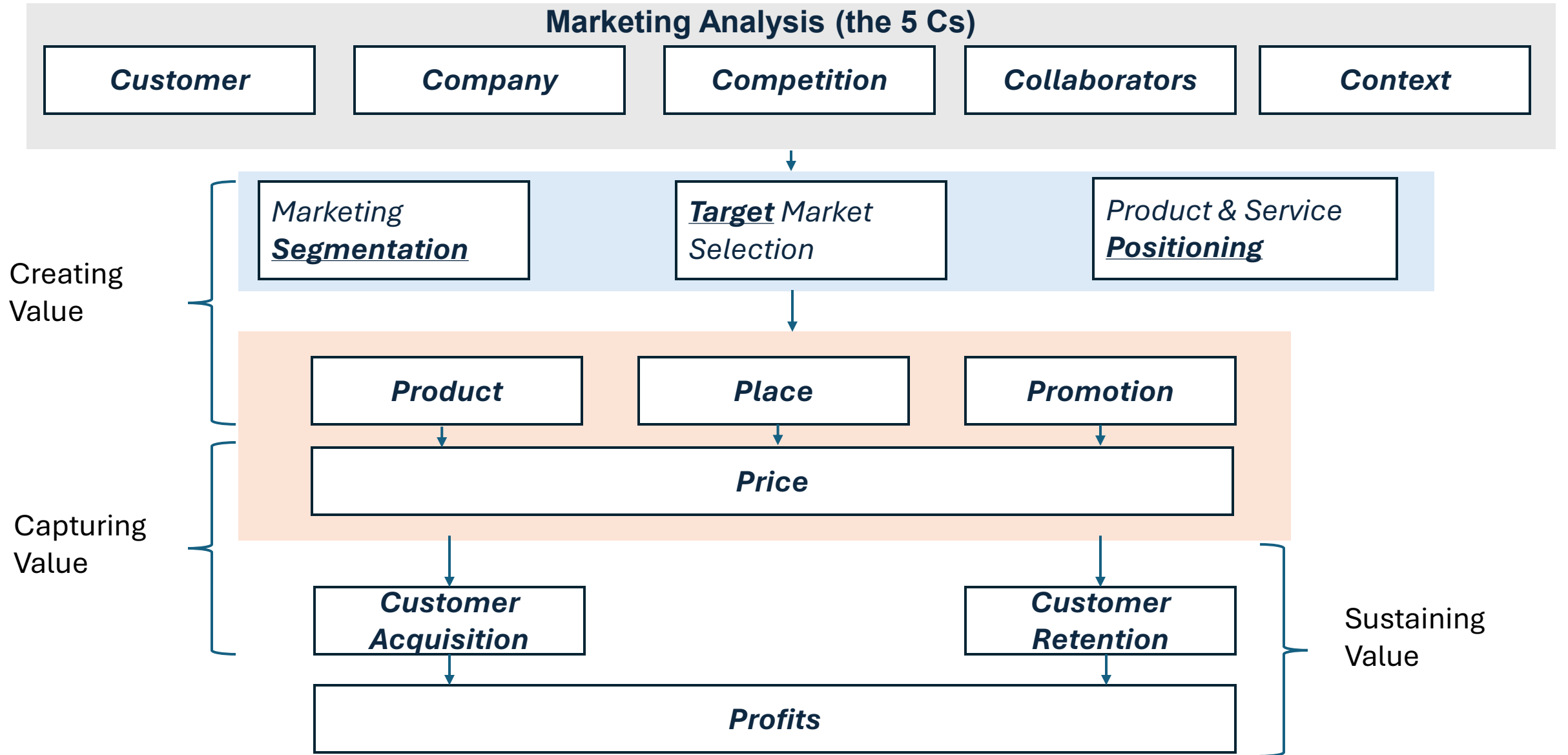


Strategy	Purpose
5Cs	Understand the market
STP	Decide who to serve and how to position
4Ps	Deliver that positioning

Btw a
mistake can
also kill a
perfectly
sound
product



Summarizing Marketing 101



My 4P

P	My Career Equivalent
Product	My technical expertise, analytical problem-solving, and ability to communicate technical ideas clearly
Price	Little more than entry-level salary expectations balanced with willingness to learn and grow
Place	Campus placements + Alumni connections
Promotion	• Resume that communicated my positioning clearly with value to the organization • Interviews • Elevator Pitch

Can we all take 15 min and create following for our individual selves

1. The 5 C
2. STP
3. 4P

Can you guess the brand?



Can you guess the brand?



McDonald's

Let's try this one....



Let's try this one....

The image shows the Coca-Cola logo, which consists of the words "Coca Cola" in a white, bold, sans-serif font. The text is centered within a white rectangular box that has a thin blue border. This box is positioned in the center of the image, overlapping two large red rectangular areas. The red areas are separated by a small white gap, and the overall composition is clean and minimalist.

Coca Cola

Last one...



Last one...





What are brands

*A name, term, design, symbol, or a combination of them, intended to **identify** the goods or services of one seller or group of sellers and to **differentiate** them from competitors*

Subtle

Subconscious

Long Term

Promise

Brand builds recall, trust and promise

PRODUCTS	BRANDS
Occupies functional territory. It does something	Brand exists in the head. It stands for something
Product is based on something tangible	Brand is based on associations. It makes you feel something
Product expands choice: 'where should we go for lunch'?	Brands simplify choice: Lets go to Subway
Products can be identical	A brand differentiates

Why Brands

Brands add value to business

By extending the **brand's penetration** to new customers in the same product category

Colgate to Colgate Total to
Colgate Sensitive to Colgate
Sensitive Pro Relief

By enabling the brand to command a **price premium** versus the competition

Tommy Hilfiger

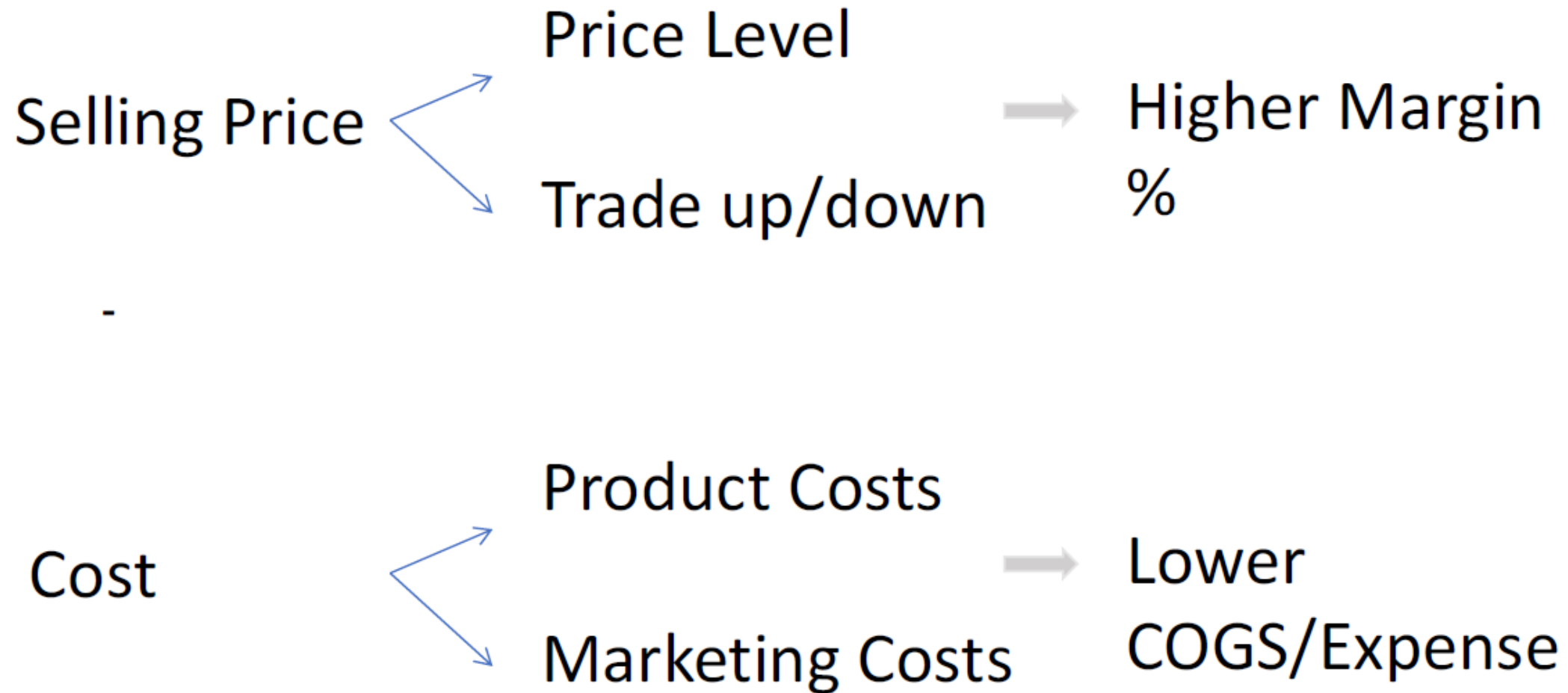
By allowing the brand to capture a **majority of sales** at a more modest price point

Ikea

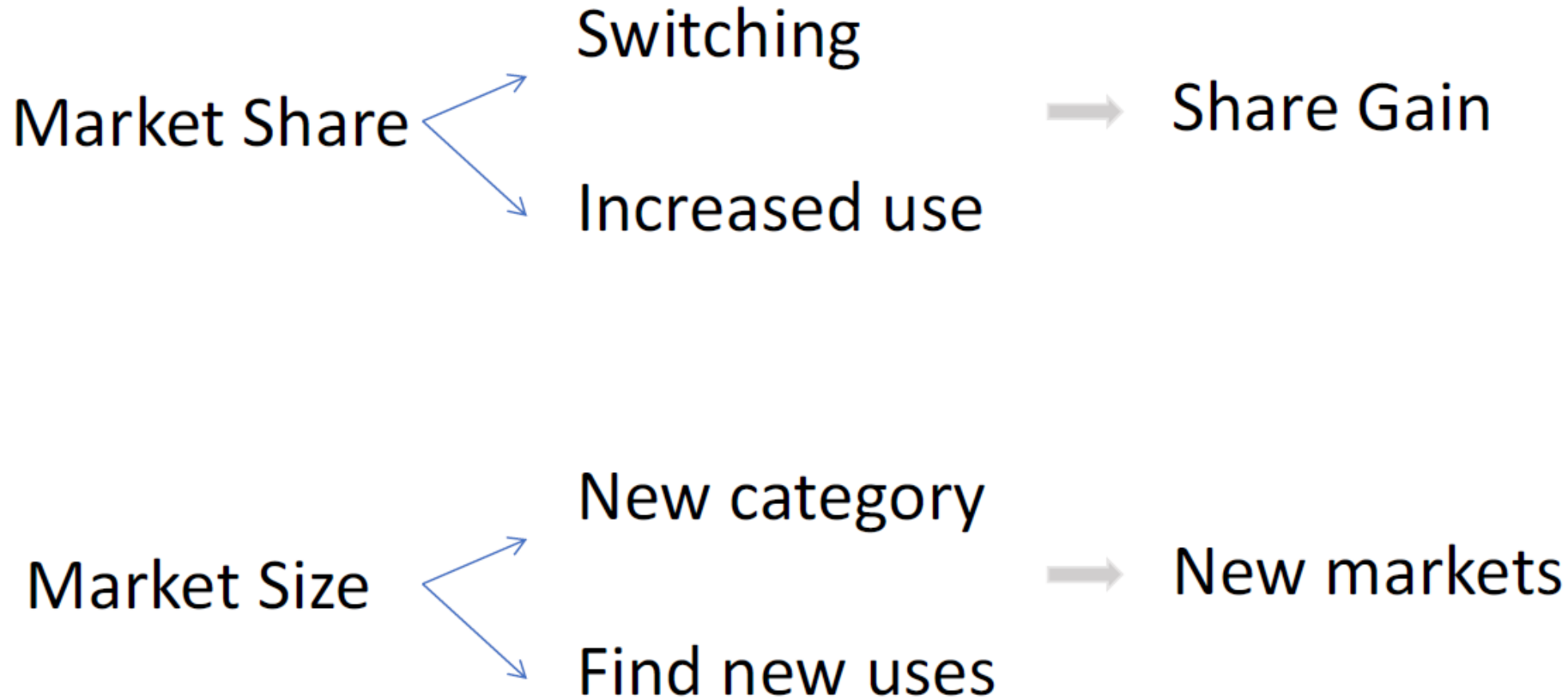
By **extending the brand's** reach to new countries and categories

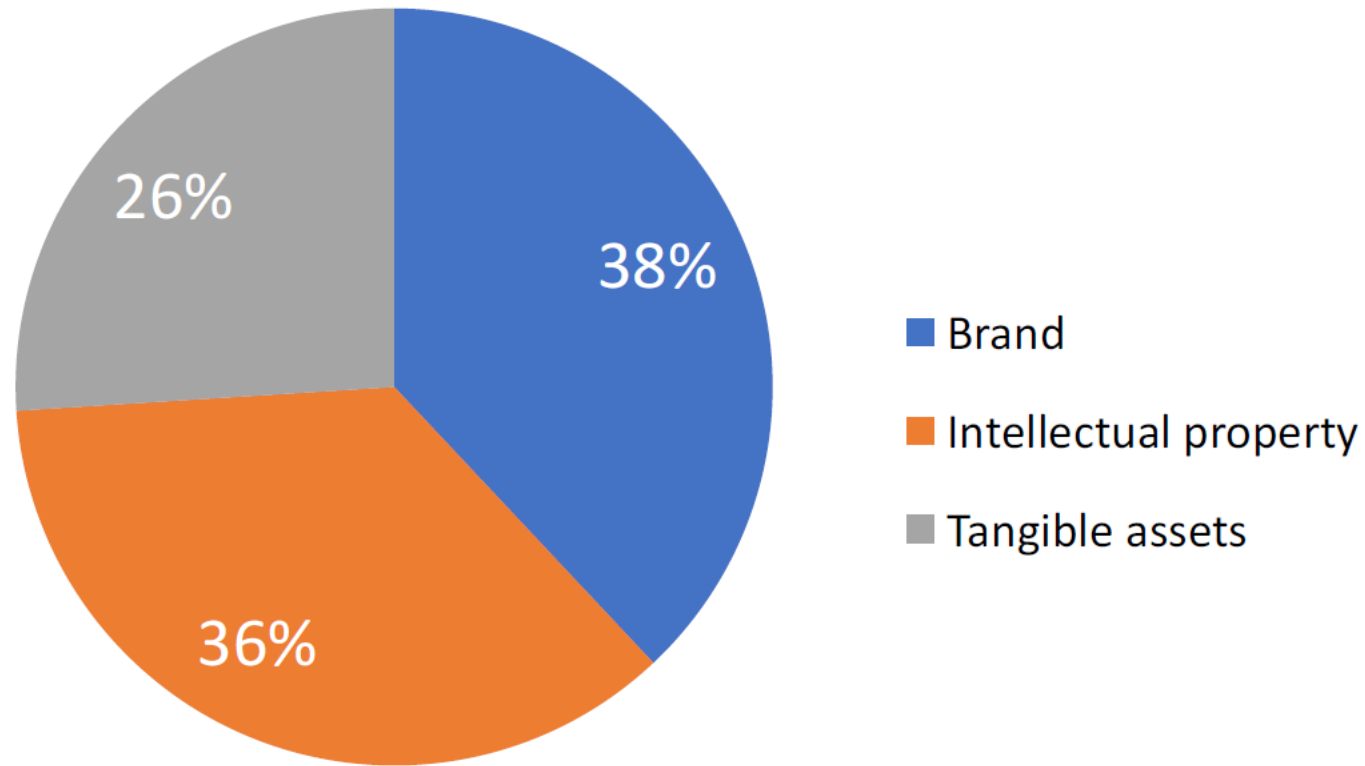
Amazon

Brands contribute to profit



Brands contribute to business





Average components of enterprise value: Top 200 companies in the world

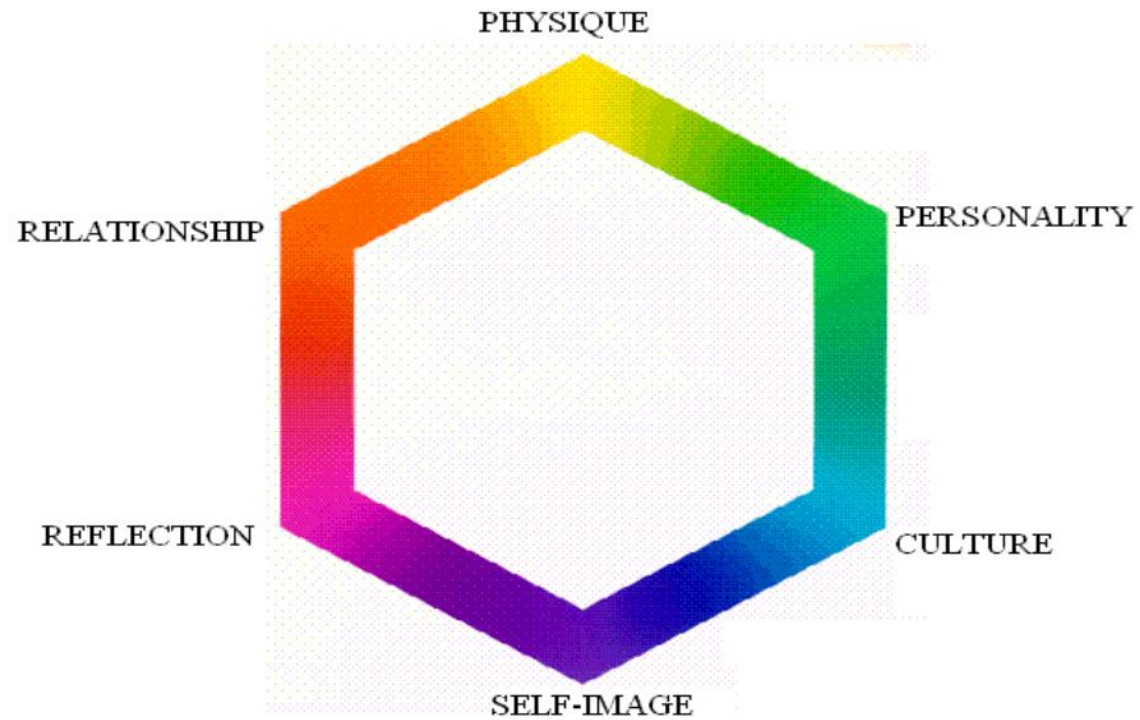
Business Week League Table 2013

Brand Identity

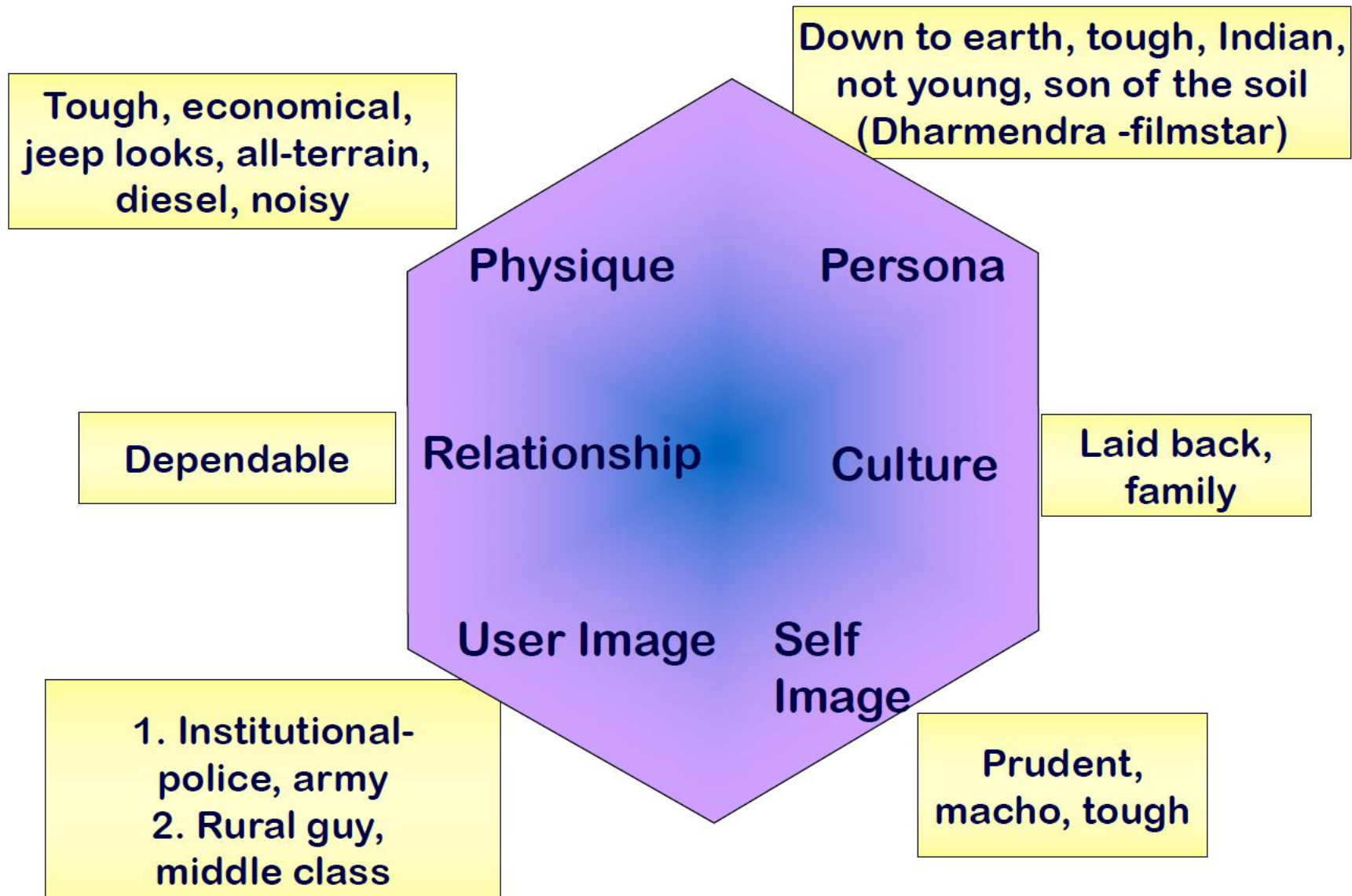
Brand Identity is the unique set of brand associations that the brand strategist aspires to create or maintain.

These associations represent what the brand stands for (or what the company wants the brand to stand for)

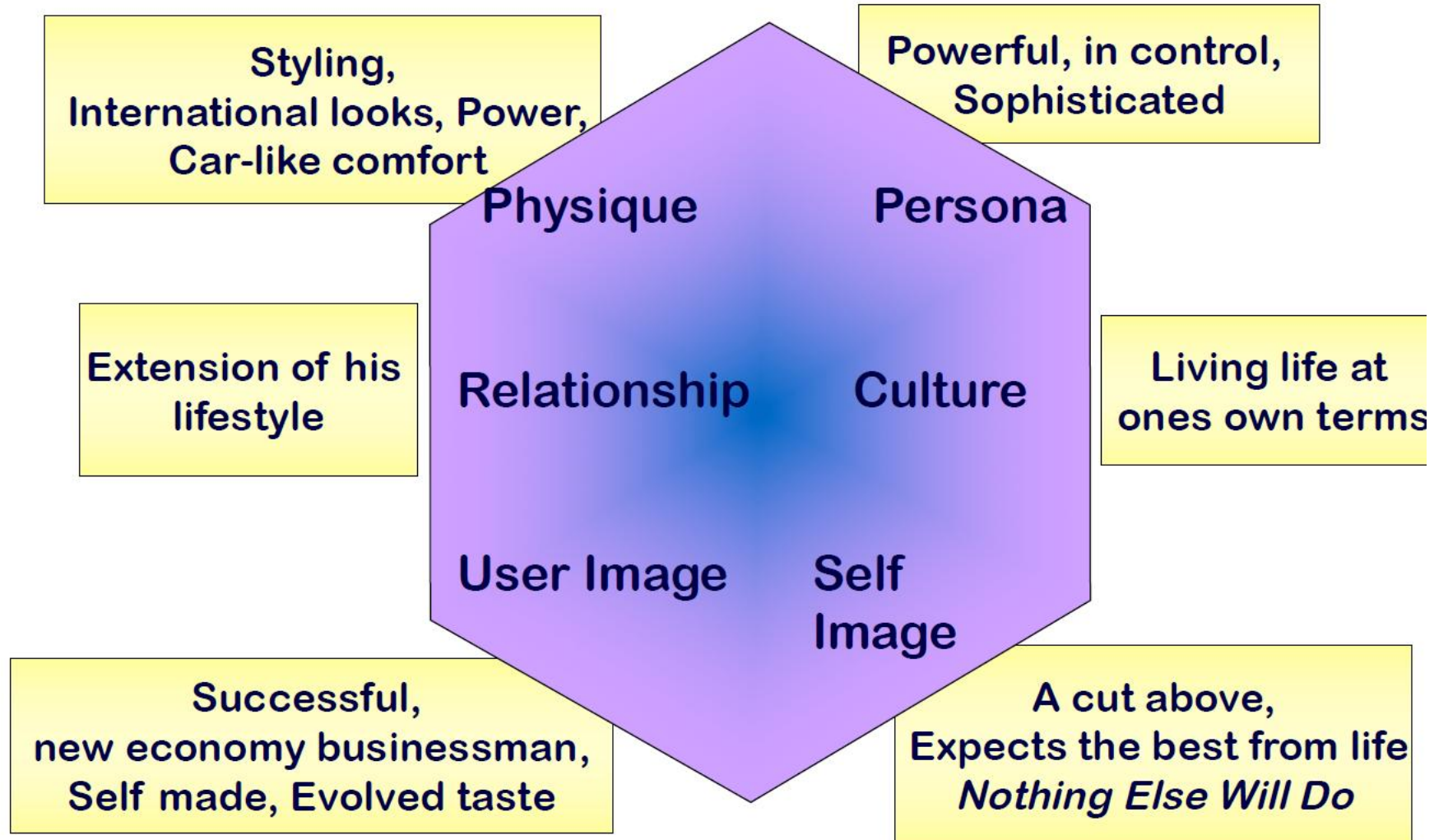
Revisiting
**KAPFERER'S FRAMEWORK
BRAND IDENTITY PRISM**



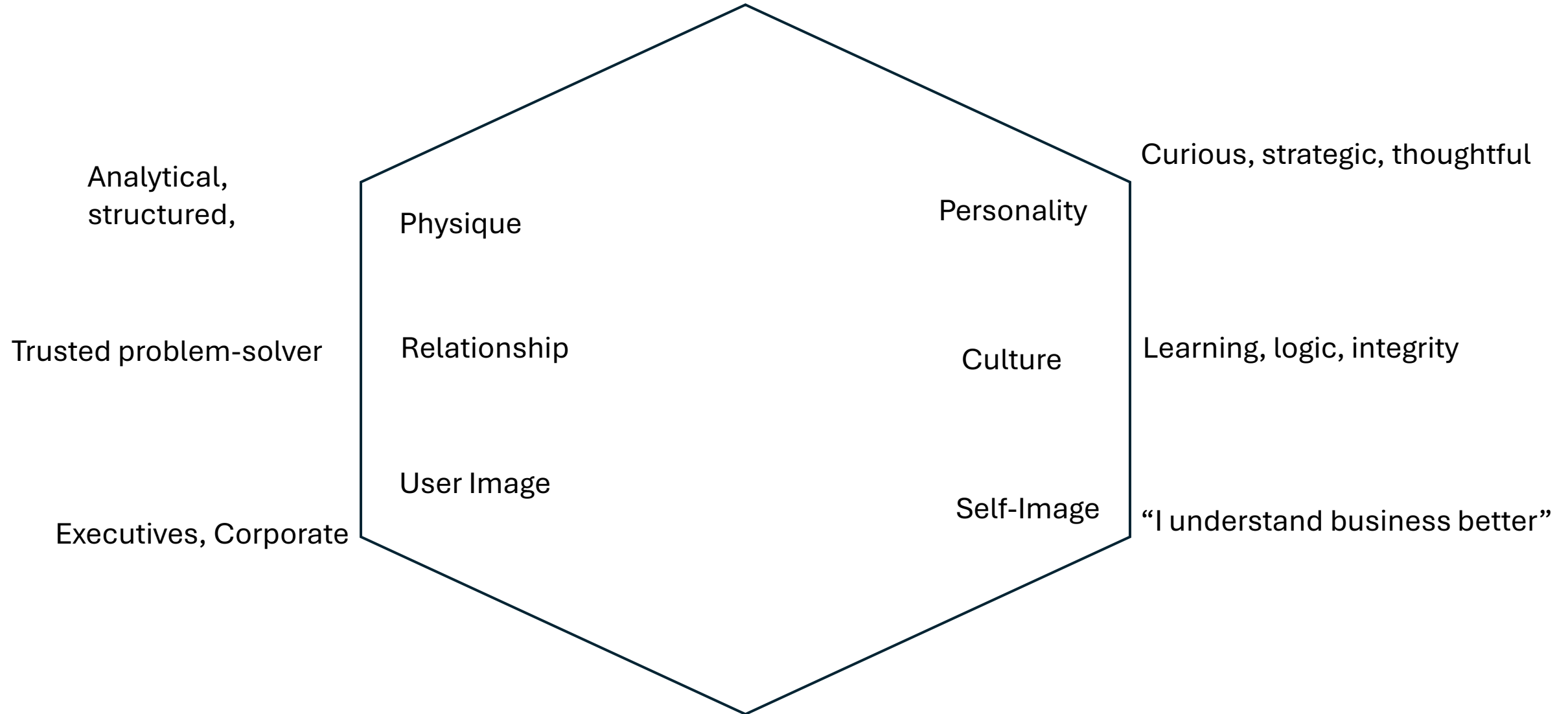
Mahindra » brand identity prism



Scorpio »



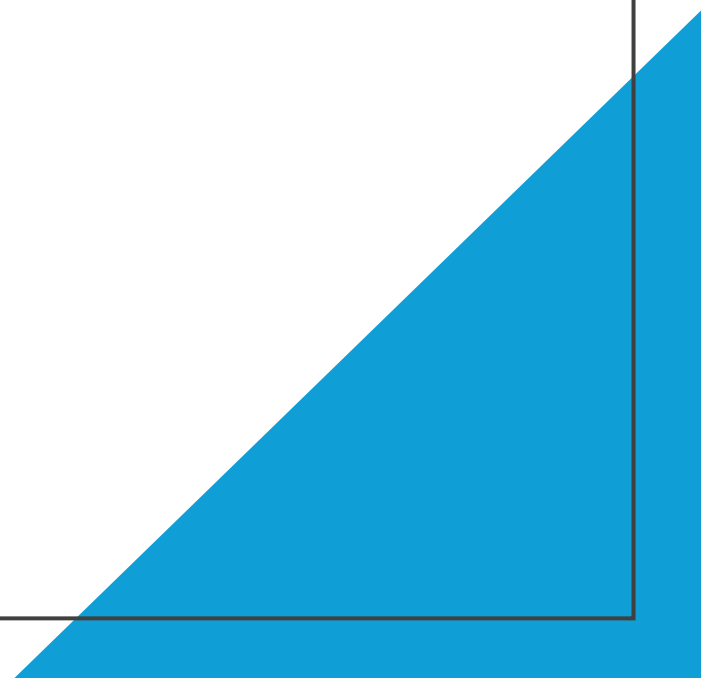
Lets create this in our groups



Key Learning

- Brands are experiences and hence build on emotions
- Brands are what its customers associate it with
- Brand drive profits and add to business value
- Brands like humans have a Soul, Body and are governed by Karma theory
- Strategic coherence is as important than execution consistency
- Brand Identity- the starting point for building brands

Can people be brands



Sales Vs Marketing



Marketing → Creates demand
Sales → Converts demand into revenue



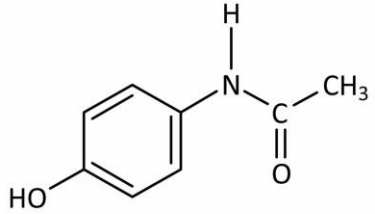
Sales can be divided into two types

B2C

(Business → Consumer)

B2B

(Business → Business)



ACETAMINOPHEN

N-acetyl-p-aminophenol

Paracetamol



A company that does both B2B and B2C ?



Bringing it all together, what did we learn today

Thank you